

Line Integrals

$$\iiint_C f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \quad (1)$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C Pdx + Qdy + Rdz \quad (\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}) \quad (2)$$

Green Theory

If $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$: The path doesn't change the value:

$$\int_C P dx + Q dy = \iint_{C_{closed}} \left(\frac{\partial Q}{\partial y} - \frac{\partial P}{\partial x} \right) dx dy$$

Stoke's formula

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}$$

$$\oint_C Pdx + Qdy + Rdz = \iint_S \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z} \right) dydz + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x} \right) dzdx + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$$